CHIRANJIV

GENOMICS

Whole Genome Sequencing Analysis Report

Disclaimer: All personal information that you see for anyone is fictitious (for privacy reasons). This report should be taken only as a Sample report and the actual nature and level of detail may vary from one individual to another

Report Details

LABORATORY
Chiranjiv

REPORT ID
CHG-WGS-2025-001847

TEST NAME
Comprehensive Whole Genome Sequencing with Phenotype Analysis

TEST CODE
WGS-COMPLETE-v3

REPORT DATE
November 26, 2025

AUTHORIZED BY
[redacted]

Patient Information

PATIENT NAME
[redacted]

DATE OF BIRTH
[redacted]

AGE
[redacted]

GENDER
Male

SAMPLE ID
CHG-SAM-2025-087342

SAMPLE COLLECTION DATE
[redacted]

SAMPLE TYPE
Whole Blood (EDTA)

DNA CONCENTRATION
156 ng/μL

DNA QUALITY (A260/A280)
1.89

Technical Summary

- Sequencing Platform: Illumina NovaSeq 6000
- Sequencing Depth: 50× (WGS)
- Read Length: 150 bp paired-end
- Total Reads Generated: 3.2 billion
- Genome Coverage: 99.9% at >10× depth
- Total Variants Detected: 4,287,342 SNPs; 687,450 Indels; 89 CNVs

Quality Metrics: Pass (Q-score >30 for 99.8% of bases)
Alignment Reference: GRCh38/hg38
Variant Calling: GATK Best Practices Pipeline v4.2

1. Wellness and Longevity Parameters

Overview

This section analyzes genetic markers associated with healthy aging, lifespan, cardiovascular health, metabolic resilience, and disease resistance. The variants identified represent common genetic variations affecting longevity pathways based on large-scale epidemiological studies.

Key Findings

1.1 Apolipoprotein E (APOE) - Aging and Cognitive Health

Gene: APOE (Chromosome 19)
Variants Analyzed: rs429358 (ε2/ε3/ε4 isoforms), rs7412

MARKER	GENOTYPE	ALLELES	ZYGOSITY	RISK LEVEL	CLINICAL SIGNIFICANCE
APOE ε3/ε4	C/T + C/C	ε3/ε4	Heterozygous	Moderate	Increased Alzheimer's disease risk (3.2-fold); reduced longevity odds (~19% lower survival to age 85); associated with accelerated physical aging
rs429358	C/T	C/T	Heterozygous	Moderate	ε4 allele carrier; cholesterol metabolism affected
rs7412	C/C	C/C	Homozygous	Lower	ε3/ε3 background provides cognitive protection

Recommendation: Regular cognitive screening starting age 50; cardiovascular risk reduction; increased omega-3 intake; cognitive engagement; consider apoE4-specific preventive strategies[1][2].

1.2 Serum Levels of Lipids and Longevity (SORT1, LDLR, APOB)

Gene: SORT1 (rs646776), LDLR (rs6511720), APOB (rs693)
Chromosome: 1, 19, 2

MARKER	GENOTYPE	ALLELES	EFFECT	LONGEVITY IMPACT
SORT1 rs646776	G/G	G/G	Homozygous	Favorable - Lower LDL cholesterol; reduced cardiovascular mortality risk
LDLR rs6511720	A/G	A/G	Heterozygous	Neutral - Moderate LDL receptor activity; balanced lipid processing
APOB rs693	A/A	A/A	Homozygous	Favorable - Lower apolipoprotein B levels; reduced atherosclerotic burden

Summary: Overall favorable lipid profile genetically; LDL cholesterol predicted to be 15-20% lower than population average.

1.3 Telomere Length and Cellular Aging (TERC, TERT)

Gene: TERC (rs12696304), TERT (rs2242652, rs7725391)
Chromosome: 3, 5

MARKER	GENOTYPE	PREDICTED PHENOTYPE	BIOLOGICAL AGE IMPACT
TERC rs12696304	C/C	Longer telomeres	+3.2 years biological youth advantage
TERT rs2242652	G/G	Favorable telomerase activity	Preserved cellular replication capacity
TERT rs7725391	C/T	Heterozygous advantage	Optimal telomere maintenance

Clinical Interpretation: Genetic predisposition toward longer telomeres and slower cellular aging. Biological age approximately 3 years younger than chronological age based on these markers[3].

1.4 Inflammation and Immune Aging (IL-6, TNF-α, CRP variants)

Gene: IL6 (rs1800795), TNF (rs1799724), CRP (rs2794520)

MARKER	GENOTYPE	BASELINE LEVEL PREDICTION	INTERPRETATION
IL6 rs1800795	G/G	Lower IL-6	Favorable - Reduced chronic inflammation; better immune aging
TNF rs1799724	G/A	Moderate TNF-α	Balanced pro-inflammatory response
CRP rs2794520	C/C	Lower CRP	Protective - Reduced cardiovascular and metabolic inflammation

Summary: Genetic profile suggests predisposition toward lower chronic inflammation—protective for healthy aging and longevity.

1.5 NAD+ Metabolism and Energy Production (NAMPT, SIRT1)

Gene: NAMPT (rs1319501), SIRT1 (rs3818292)

MARKER	STATUS	FUNCTIONAL IMPACT
NAMPT rs1319501	Heterozygous	Normal NAD+ biosynthesis pathway; preserved mitochondrial function
SIRT1 rs3818292	Homozygous wild-type	Optimal sirtuin activity; enhanced stress resistance

Implication: Efficient NAD+ metabolism supports cellular energy production and senescence suppression—favorable for longevity.

Wellness Summary Score

Longevity Index: 72/100 (Above average)

- Favorable lipid genetics (+12 points)
- Longer telomere predisposition (+10 points)
- Lower inflammation profile (+15 points)
- APOE ε4 carrier status (-5 points)
- Optimal NAD+ metabolism (+8 points)
- Normal metabolic flexibility (+12 points)

2. Ancestry and Population Genetics Insights

Overview

This section determines genetic ancestry composition, continental origin, population substructure, and migration history based on genome-wide SNP patterns and principal component analysis.

Ancestry Composition

Primary Ancestry: South Asian (92.4%)
Secondary Ancestry: Central Asian (5.3%)
Tertiary Ancestry: Mediterranean (2.3%)
Confidence Level: 99.1%

2.1 Continental Ancestry Breakdown

CONTINENTAL REGION	PERCENTAGE	CONFIDENCE
South Asia (Indian subcontinent)	92.4%	99.1%
Central Asia	5.3%	95.2%
Mediterranean/Middle East	2.3%	87.4%

Interpretation: Ancestry profile consistent with individuals from Northern India, likely with roots in Punjab, Haryana, or Himachal Pradesh regions. Central Asian ancestry suggests possible admixture from historical Indo-European migrations (~2000-1500 BCE).

2.2 Subcontinental South Asian Breakdown

SUBCLADE	PERCENTAGE	GEOGRAPHIC REGION
Indo-Aryan (North Indian)	78.1%	Northern India, Upper Gangetic Plain
Dravidian admixture	12.8%	South Indian ancestral component
Austro-Asiatic	1.5%	Eastern India ancestral component

2.3 Y-Chromosome Haplogroup (Male-Specific)

Y-Haplogroup: R1a1a (M198+, M417+)
Subclade: R1a1a-Z284 (Z282+, Z283+)
Geographic Origin: Central Asia to South Asia (Indo-European expansion)
Estimated Age of Clade: ~4,500 years before present
Migration Pattern: Associated with Indo-European speaker spread into South Asia

Interpretation: Y-chromosome haplogroup R1a1a is predominant in Northern India and associated with Indo-European languages. This lineage shows evidence of expansion during Bronze Age migrations (~2000-1500 BCE).

2.4 Mitochondrial DNA Haplogroup (Maternal Lineage)

mtDNA Haplogroup: M (ancestral South Asian clade)
Subclade: M5a
Geographic Origin: South Asia
Estimated Age: ~8,000-12,000 years before present

Interpretation: Maternal lineage represents early South Asian settlement, predating or contemporary with Indo-European migrations. This haplogroup is particularly common in North and Central India.

2.5 Genetic Distance to Global Populations

POPULATION	GENETIC DISTANCE	SIMILARITY
North Indian	0.012	Very Close
Punjabi	0.018	Very Close
Pakistani	0.021	Close
Central Asian	0.087	Moderate
European	0.156	Distant
East Asian	0.198	Very Distant

POPULATION	GENETIC DISTANCE	SIMILARITY
African	0.213	Very Distant

2.6 Disease Allele Frequency Comparisons

Your genome was compared to disease allele frequencies in your ancestral populations vs. European reference populations:

VARIANT	YOUR STATUS	SOUTH ASIAN FREQ	EUROPEAN FREQ	IMPLICATION
FTO rs9939609	Heterozygous (A/T)	48.2%	42.3%	Common in your ancestral population
APOE ε4	Heterozygous	22.1%	18.7%	Slightly enriched in South Asian ancestry
MTHFR C677T	Heterozygous	31.5%	28.2%	Common variant across populations

Key Ancestry Findings

- Confirmed South Asian heritage with high confidence
- Subcontinental origin: Northern India (likely Punjab/Haryana region)
- Maternal lineage: Ancient South Asian settlement
- Paternal lineage: Indo-European expansion (Bronze Age)
- No evidence of: Sub-Saharan African, East Asian, or Ashkenazi Jewish ancestry
- Expected genetic diversity: Consistent with North Indian/Punjabi populations

3. Predisposition to Lifestyle and Metabolic Disorders

Overview

This section evaluates genetic susceptibility to conditions influenced by lifestyle factors including obesity, type 2 diabetes, hypertension, and dyslipidemia. These multifactorial traits result from gene-environment interactions.

3.1 Obesity and Weight Management

FTO (Fat Mass and Obesity-Associated) Gene

Gene: FTO (rs9939609, rs1421085)
Chromosome: 16
Variant Function: Intronic SNP affecting FTO expression in hypothalamus

MARKER	GENOTYPE	ALLELES	RISK ALLELE COPIES	RISK LEVEL	EFFECT SIZE
FTO rs9939609	A/T	rs9939609	1 copy	Moderate	+1.2 kg/m² BMI per risk allele
FTO rs1421085	C/G	rs1421085	1 copy	Moderate	+0.9 kg/m² BMI per risk allele

Functional Impact:

- FTO A allele (risk) associated with increased appetite and food-seeking behavior
- Altered expression affects leptin signaling in ventromedial hypothalamus
- Increased preference for energy-dense, high-fat foods
- ~1 kg higher body weight expected per copy of risk allele

Predicted Phenotype: Moderate genetic predisposition to weight gain; approximately 3-4 kg above-average baseline weight expectation attributable to these variants alone[4].

Mitigating Strategies:

- Regular structured exercise (150+ min/week moderate intensity)
- Portion control and mindful eating
- Reduced consumption of high-fat, energy-dense foods
- Increased protein intake (improves satiety)
- Intermittent fasting or time-restricted eating protocols may be particularly effective for A allele carriers

MC4R (Melanocortin-4 Receptor) Gene

Gene: MC4R (rs17782313, rs52820871)
Chromosome: 18

MARKER	GENOTYPE	STATUS	RISK IMPACT
rs17782313	C/T	Heterozygous	Modestly increased obesity risk (1.08 OR)

MARKER	GENOTYPE	STATUS	RISK IMPACT
rs52820871	T/T	Homozygous	Low-risk genotype for severe obesity

Clinical Significance: MC4R variants associated with monogenic obesity when homozygous; your heterozygous status contributes modestly to population-level obesity risk.

3.2 Type 2 Diabetes Mellitus (T2DM)

TCF7L2 (Transcription Factor 7-Like 2) Gene

Gene: TCF7L2 (rs7903146, rs12255372)
Chromosome: 10
Strongest GWAS Signal for T2DM

MARKER	GENOTYPE	RISK ALLELE COPIES	RELATIVE RISK	MECHANISM
TCF7L2 rs7903146	C/T	1 copy	1.42× increased risk	Impaired β-cell function; reduced GLP-1 response
TCF7L2 rs12255372	A/G	1 copy	1.39× increased risk	Wnt/β-catenin pathway dysregulation

Predicted T2DM Risk: 16.3% lifetime risk (population baseline ~12.5%) — approximately 1.31× population average

Clinical Recommendation:

- Fasting glucose and HbA1c screening annually (or biannually post-age 45)
- Oral glucose tolerance test every 2-3 years
- Lifestyle interventions emphasize 7% weight loss if overweight
- Mediterranean or low-glycemic-index diets recommended

KCNJ11 (Potassium Channel Gene) and SLC30A8 (Zinc Transporter)

GENE	SNP	GENOTYPE	EFFECT
KCNJ11	rs5219	C/C	1.09× increased T2DM risk; impaired insulin secretion

GENE	SNP	GENOTYPE	EFFECT
SLC30A8	rs13266634	C/T	1.06× increased T2DM risk; reduced β-cell zinc transport

3.3 Dyslipidemia and Lipid Metabolism

PCSK9 (Proprotein Convertase Subtilisin/Kexin Type 9)

Gene: PCSK9 (rs505151, rs11206510)
Chromosome: 1
Function: LDL receptor degradation; strong impact on LDL cholesterol

MARKER	GENOTYPE	EFFECT	LDL IMPACT
PCSK9 rs505151	G/G	Homozygous	Favorable — 15-20% lower LDL-C
PCSK9 rs11206510	T/T	Homozygous	Enhanced LDL clearance

Predicted Lipid Profile:

- LDL Cholesterol: 95-105 mg/dL (genetic prediction)
- HDL Cholesterol: 52-58 mg/dL (favorable)
- Triglycerides: 115-125 mg/dL (normal)
- Lipoprotein(a): Normal range expected

CETP (Cholesteryl Ester Transfer Protein)

Gene: CETP (rs3764261)

MARKER	GENOTYPE	EFFECT
CETP rs3764261	G/G	Higher HDL; lower triglycerides; favorable lipid profile

3.4 Hypertension Susceptibility

AGT (Angiotensinogen) Gene

Gene: AGT (rs699, M235T variant)
Chromosome: 1

MARKER	GENOTYPE	BLOOD PRESSURE EFFECT	RISK ASSESSMENT
AGT M235T	M/T	Heterozygous — neutral	1.08× increased hypertension risk

Clinical Interpretation: Mild genetic predisposition to elevated blood pressure. Combined with lifestyle factors (sodium intake, stress, sedentary behavior), increased monitoring recommended.

CYP11B2 (Aldosterone Synthase)

Gene: CYP11B2 (-344 C/T)

MARKER	GENOTYPE	EFFECT	ALDOSTERONE
CYP11B2 -344C/T	C/T	Heterozygous	Moderate aldosterone levels; normal sodium retention

3.5 Metabolic Syndrome Risk

Metabolic Syndrome Z-Score: +0.42 (slightly above average)
Risk Category: Intermediate (44th percentile)

COMPONENT	GENETIC RISK	PHENOTYPIC TARGET
Central Obesity	Moderate	Waist circumference <94 cm
Fasting Glucose	Moderate	Fasting glucose <100 mg/dL
HDL Cholesterol	Low-Moderate	HDL >40 mg/dL (males)
Triglycerides	Moderate	Triglycerides <150 mg/dL
Blood Pressure	Mild	<130/85 mmHg

3.6 Summary of Lifestyle & Metabolic Risk

CONDITION	GENETIC RISK	LIFETIME RISK	RECOMMENDATION

CONDITION	GENETIC RISK	LIFETIME RISK	RECOMMENDATION
Obesity	Moderate	34%	Weight management, exercise
Type 2 Diabetes	Moderate-High	16.3%	Annual screening, lifestyle
Dyslipidemia	Mild	22%	Dietary intervention likely sufficient
Hypertension	Mild	28%	Sodium reduction, regular exercise
Metabolic Syndrome	Moderate	31%	Holistic lifestyle approach

Overall Metabolic Health Score: 64/100 (Below average — proactive management recommended)

4. Susceptibility to Common and Complex Diseases

Overview

This section analyzes polygenic risk scores (PRS) and rare/high-penetrance variants associated with chronic diseases and complex conditions, including cardiovascular disease, cancer, neurological disorders, and autoimmune diseases.

4.1 Cardiovascular Disease

4.1.1 Coronary Artery Disease (CAD) and Myocardial Infarction (MI)

Polygenic Risk Score (PRS) for CAD: 68th percentile
Lifetime Risk (to age 75): 19.2% (population average: 16.8%)
Relative Risk: 1.14× population average

Key Risk Variants Identified:

GENE	SNP	GENOTYPE	EFFECT	MECHANISM

GENE	SNP	GENOTYPE	EFFECT	MECHANISM
CDKN2B-AS1	rs1333049	C/G	Risk allele	Cell cycle regulation; vascular smooth muscle proliferation
SORT1	rs646776	G/A	Protective	LDL cholesterol metabolism
CYP17A1	rs1893590	T/T	Risk allele	Lipid and steroid metabolism
ZEB2	rs2943634	G/G	Risk allele	Endothelial dysfunction

Protective Variants:

- SORT1 rs646776 (G/G): 12% risk reduction
- LDLR variants: 8% risk reduction

Clinical Assessment:

- 10-Year ASCVD Risk (Framingham): Estimated 2.1% (low-intermediate)
- Recommend baseline lipid panel, blood pressure monitoring
- Consider CAC (coronary artery calcium) screening at age 55 if additional risk factors present

4.1.2 Atrial Fibrillation (AFib)

PRS for AFib: 52nd percentile
Lifetime Risk (age 40-lifetime): 18.4%
Relative Risk: 1.02× population average

Key Loci:

- PITX2 (rs2637674): Heterozygous
- ZFHX3 (rs7212175): Heterozygous
- CAN (Calcium channel): Normal

Recommendation: Screening ECG at age 50; awareness of AFib symptoms; regular pulse checks recommended.

4.1.3 Stroke Risk

Ischemic Stroke PRS: 55th percentile
Lifetime Risk: 12.3%

Protective factors: No rare pathogenic variants detected in stroke-related genes (NOTCH3, COL4A1)

4.2 Metabolic Diseases

4.2.1 Type 2 Diabetes (Additional Detail)

Comprehensive T2DM PRS: 71st percentile
Lifetime Risk (age 35-lifetime): 16.3%

Additional variants beyond Section 3:

GENE	SNP	RISK EFFECT
MTNR1B	rs10830963	1.05× increased risk
PPARGC1A	rs8192678	1.08× increased risk
HHEX	rs1111875	1.06× increased risk

4.2.2 Chronic Kidney Disease (CKD)

CKD PRS: 48th percentile (lower risk)
5-Year CKD Incidence Risk: 2.1%

Key Protective Variants: UMOD (rs12917707) shows protective allele; normal APOL1 genotype (non-African ancestry)

4.3 Cancer Susceptibility

4.3.1 Colorectal Cancer (CRC)

CRC PRS: 62nd percentile
Lifetime Risk (to age 85): 6.8% (population average: 4.2%)
Relative Risk: 1.62× population average — ELEVATED

Key Risk Loci Identified:

REGION	SNP	GENOTYPE	EFFECT	OR
8q23.3	rs6983267	G/G	Homozygous risk	1.31
18q21.1	rs4939827	C/C	Homozygous risk	1.19

REGION	SNP	GENOTYPE	EFFECT	OR
10p14	rs10795668	A/A	Homozygous risk	1.21
11q23.1	rs3802842	A/A	Homozygous risk	1.24
15q13.3	rs4779584	G/G	Homozygous risk	1.15

No pathogenic mutations in: MLH1, MSH2, MSH6, PMS2 (Lynch syndrome genes)
No pathogenic mutations in: APC (familial adenomatous polyposis)

Clinical Recommendations (CRITICAL):

- Initiate colonoscopy screening at age 40 (rather than standard age 45-50)
- Screening interval: Every 7-10 years if normal
- Increase surveillance frequency if polyps detected
- Consider fecal DNA testing (Cologuard) annually as adjunct
- Lifestyle modifications:
  o Increase dietary fiber (30+ g/day)
  o Reduce processed/red meat consumption
  o Regular physical activity (150+ min/week)
  o Maintain healthy weight (BMI 18.5-24.9)
  o Avoid smoking and excessive alcohol

4.3.2 Breast Cancer (Female Relatives)

Breast Cancer PRS: 56th percentile
Familial Risk: No high-penetrance BRCA1/BRCA2 pathogenic variants detected

Key Common Risk Variants:

LOCUS	SNP	EFFECT
LSP1	rs3817116	Modest increased risk
H19	rs2107425	Modest increased risk
TOX3	rs4784227	Modest increased risk

Lifetime Risk (female relatives): 14.2% (vs. population baseline ~12.8%)
Recommendation: Standard mammography screening guidelines; no early augmented screening needed unless family history of early-onset breast cancer.

4.3.3 Prostate Cancer

Prostate Cancer PRS: 58th percentile
Lifetime Risk (to age 85): 16.1%

Key Risk Variants:

REGION	SNP	GENOTYPE	EFFECT
8q24.21	rs1447295	Heterozygous	Modest risk elevation
17q24.3	rs1859962	Heterozygous	Modest risk elevation
11q13	rs1016343	Heterozygous	Modest risk elevation

Recommendation:

- Baseline PSA at age 50
- Shared decision-making regarding screening (PSA screening offers benefit but also false positives)
- Digital rectal exam optional; low-risk genotype profile does not mandate aggressive screening

4.3.4 Lung Cancer

Lung Cancer PRS: 44th percentile (LOWER than average)
Lifetime Risk (non-smokers): 1.2%

Status: No significant genetic risk identified. Smoking status is the primary determinant of risk.

4.4 Neurological Diseases

4.4.1 Alzheimer's Disease (AD)

AD PRS: 71st percentile — ELEVATED RISK
Lifetime Risk (to age 85): 14.7% (vs. population baseline: 9.1%)
Relative Risk: 1.62× population average

Key Genetic Contributions:

GENE	SNP	GENOTYPE	EFFECT	RISK CONTRIBUTION
APOE	ε3/ε4	Heterozygous	Strong risk	3.2× increased risk for AD[1]

GENE	SNP	GENOTYPE	EFFECT	RISK CONTRIBUTION
BIN1	rs744373	G/G	Risk allele	1.12× increased risk
CLU	rs11136000	C/C	Protective	0.88× (slight protection)
CR1	rs3818361	A/A	Risk allele	1.13× increased risk
PICALM	rs3851179	A/A	Risk allele	1.11× increased risk

Composite Assessment: APOE ε4 heterozygosity is the dominant risk factor; combined with elevated PRS across multiple loci.

Risk Stratification:

- Age 50-65: Preclinical/asymptomatic stage; cognitive decline risk increases
- Age 65-75: Mild cognitive impairment risk window
- Age >75: Clinical dementia risk increases exponentially

Recommended Interventions (Evidence-Based):

- Cognitive training: Dual n-back training, learning new skills
- Cardiovascular health: Maintain blood pressure <130/80 mmHg; control dyslipidemia
- Physical exercise: 150+ min/week aerobic activity (supports BDNF); resistance training 2×/week
- Cognitive reserve: Pursue higher education/intellectual engagement; bilingualism (beneficial—patient speaks English and Hindi)
- Sleep optimization: 7-9 hours nightly; manage sleep apnea if present
- Diet: Mediterranean or MIND diet (strong AD protection data)
- Inflammation reduction: Omega-3 supplementation; limit processed foods
- Monitor cognitive function: Baseline cognitive assessment at age 60; repeat every 2 years
- Biomarker monitoring: Consider amyloid PET or blood biomarkers (phospho-tau, p-tau181) at age 65-70 if available

Note on Intervention: Recent anti-amyloid monoclonal antibodies (aducanumab, lecanemab) show modest slowing of cognitive decline in APOE ε4 carriers with mild cognitive impairment; discuss eligibility if applicable.

4.4.2 Parkinson's Disease

Parkinson's Disease PRS: 52nd percentile
Lifetime Risk: 2.8%

Status: No high-risk variants identified; standard population risk applies.

4.4.3 Multiple Sclerosis (MS)

MS PRS: 41st percentile — LOWER than average
Lifetime Risk: 0.34%
Status: No significant genetic susceptibility identified; protective genotypes at HLA-DRA and other MS loci present.

4.5 Autoimmune and Inflammatory Diseases

4.5.1 Rheumatoid Arthritis (RA)

RA PRS: 51st percentile
Lifetime Risk: 1.8%

HLA Status: HLA-DRB1 genotype: DRB101:01/DRB104:05
- Low-moderate shared epitope (SE) presence
- Moderate RA genetic risk contribution

Clinical Note: RF (rheumatoid factor) and anti-CCP status unknown; would require serology.

4.5.2 Type 1 Diabetes

T1DM PRS: 38th percentile — LOWER than average
Lifetime Risk: 0.9%

HLA Status: HLA-DR3 or HLA-DR4 genotype determines risk; your profile shows lower-risk HLA haplotypes.

4.5.3 Celiac Disease

Celiac PRS: 44th percentile
Lifetime Risk: 3.2%

HLA-DQ2/DQ8 Status: Negative for both DQ2 and DQ8 heterodimers
Clinical Interpretation: Very low risk; celiac disease unlikely even with gluten exposure.

4.6 Summary Risk Matrix

DISEASE CATEGORY	CONDITION	RISK LEVEL	PERCENTILE	ACTION

DISEASE CATEGORY	CONDITION	RISK LEVEL	PERCENTILE	ACTION
Cancer	Colorectal Cancer	Elevated	62nd	Early colonoscopy age 40; frequent screening
	Prostate Cancer	Average	58th	Shared decision-making; baseline PSA age 50
	Breast Cancer (relatives)	Average	56th	Standard screening guidelines
Neurological	Alzheimer's Disease	Elevated	71st	Cognitive monitoring; intensive prevention
	Parkinson's Disease	Average	52nd	Standard population risk
	Multiple Sclerosis	Low	41st	Very low risk
Cardiovascular	Coronary Artery Disease	Mild-Moderate	68th	Standard ASCVD screening
	Atrial Fibrillation	Average	52nd	Age 50 ECG screening
Metabolic	Type 2 Diabetes	Moderate-High	71st	Annual screening; intensive lifestyle
	Chronic Kidney Disease	Low	48th	Standard screening
Autoimmune	Rheumatoid Arthritis	Average	51st	Standard screening
	Type 1 Diabetes	Low	38th	Very low risk
	Celiac Disease	Very Low	N/A	Negative HLA; no risk

5. Genetic Factors Related to Career/Professional Fit

Overview

This emerging field examines genetic contributions to personality traits, cognitive abilities, and aptitudes that influence occupational performance and career satisfaction. These include traits like spatial reasoning, conscientiousness, verbal intelligence, risk tolerance, and neurobiological traits affecting career domains.

Important Note: Genetic predisposition is one factor among many (education, experience, motivation, opportunity, environmental factors). These findings are exploratory and should not be used for career-limiting decisions.

5.1 Cognitive Abilities and Learning

5.1.1 General Cognitive Ability (g-factor)

Polygenic Score for Educational Attainment: 58th percentile
Predicted IQ Range: 100-108 (average to high-average)
Learning Style: Mixed profile (analytical and creative strengths)

Key Cognitive Variants:

GENE/LOCUS	SNP	GENOTYPE	COGNITIVE DOMAIN	EFFECT
HMGN1	rs1990622	G/G	General cognition	Favorable
DCDC1	rs807701	G/A	Reading ability	Heterozygous
DCAF4L2	rs11252127	A/A	Working memory	Favorable

Career Implications:

- Suited for roles requiring: Analytical thinking, problem-solving, academic learning
- Learning approach: Structured learning environments; benefits from written materials and verbal instruction
- Career trajectories: Professional, technical, management roles with clear advancement pathways

5.1.2 Verbal Intelligence and Language

Language Processing PRS: 62nd percentile
Verbal Reasoning: Above-average capacity

Supporting Evidence:
● Bilingual capacity (English and Hindi) — genetic predisposition toward language aptitude confirmed by phenotype
● Genetic variants in language-related genes (CNTNAP2, DCDC1) show favorable alleles

Career Implications:

- Communication roles: Strong candidate for roles requiring written/verbal communication
- Technical writing: Documentation, research communication
- Bilingual advantage: Career opportunities in international settings or bilingual contexts

5.1.3 Spatial Reasoning and Visuospatial Processing

Spatial Reasoning PRS: 51st percentile (average)

Key Loci:
● ROBO1 (rs3794764): Heterozygous — average spatial ability
● DLG4 (rs17178082): Normal genotype

Career Implications:

- Moderate fit for: Engineering (non-highly technical branches), architecture, design
- Better fit for: Verbal/analytical roles than highly spatial-dependent fields (e.g., neurosurgery, advanced CAD)

5.2 Personality Traits and Behavioral Characteristics

5.2.1 Conscientiousness and Diligence

Conscientiousness PRS: 68th percentile — ABOVE AVERAGE

Genetic Basis:
● rs4680 (COMT Val158Met): Val/Met heterozygous — optimal dopamine regulation
● rs1360780 (FKBP5): Normal alleles — good stress resilience

Phenotypic Expression:
● Organized, reliable, responsible work approach
● Good follow-through on projects

● Detail-oriented; quality-focused

Career Strengths:

- Suited for: Project management, quality assurance, regulatory compliance, research
- Leadership style: Organized, systematic, accountability-focused
- Career longevity: High conscientiousness associated with career stability and advancement

5.2.2 Openness to Experience and Creativity

Openness PRS: 56th percentile (average-to-above-average)

Genetic Contributions:
● DRD4 (dopamine receptor): 7R allele (heterozygous) — moderate novelty-seeking
● 5-HTTLPR (serotonin transporter): Long/short combination — balanced emotional processing

Phenotypic Expression:
● Balanced between preference for routine and openness to new ideas
● Creative capacity present but not dominant trait
● Adaptable to new situations

Career Implications:

- Well-suited for: Roles requiring both structure and innovation (e.g., biomedical research, healthcare leadership)
- Optimal roles: Integrating established protocols with innovative problem-solving
- Less optimal for: Highly creative, artistic fields requiring dominant novelty-seeking

5.2.3 Extraversion and Social Engagement

Extraversion PRS: 44th percentile — INTROVERTED tendency

Genetic Basis:
● DRD4 exon III (VNTR): 4R genotype (homozygous) — lower novelty-seeking in social context
● Amygdala reactivity genes (CRHR1): Normal alleles

Career Implications:

- Strengths: Deep work, focus, careful analysis, one-on-one relationships
- Potential challenges: Large group presentations, highly socially-demanding roles
- Optimal roles: Research scientist, data analyst, clinical specialist, technical expert

● Development: Social skills trainable; introversion not a career limitation but should match role demands
Note: Introversion in biomedical research is often advantageous for focused laboratory work.

5.2.4 Neuroticism and Emotional Stability

Emotional Stability PRS: 59th percentile (average)

Key Variants:
● CRHR1 (stress reactivity): Normal alleles
● 5-HTTLPR (serotonin transporter): LS genotype — balanced serotonergic tone

Stress Response: Moderate stress reactivity; manageable with appropriate coping strategies

Career Implications:

- Good fit for: Healthcare roles (moderate emotional demands)
- Requires: Stress management practices (exercise, mindfulness, work-life balance)
- Avoid: Roles with chronic high stress without support systems

5.3 Risk Tolerance and Decision-Making

5.3.1 Financial and Professional Risk Tolerance

Risk Tolerance Genetic Score: 52nd percentile (average-to-slightly-conservative)

Genetic Contributions:
● COMT Val158Met (dopamine): Val/Met — moderate dopamine; balanced approach to risk
● MAOA (monoamine oxidase): Normal activity

Career Implications:

- Decision-making style: Cautious but not risk-averse; prefers data-driven decisions
- Career trajectory: Stable organizational growth preferred over high-risk startup environment
- Entrepreneurship: Less suited for highly speculative ventures; better for calculated business initiatives
- Research: Good fit for hypothesis-driven research with peer review; less suited for highly speculative research domains

5.3.2 Approach-Motivation (BAS) vs. Avoidance-Motivation (BIS)

Approach/Avoidance Balance: Balanced (BAS=BIS)
Implication: Ability to weigh opportunities and threats equally; good for risk assessment roles

5.4 Occupational Phenotype Summary

Recommended Career Domains (Best Fit):

DOMAIN	FIT LEVEL	RATIONALE
Biomedical Research	Excellent	Analytical cognition, conscientiousness, focus suitable; introversion advantage
Genomics/Bioinformatics	Excellent	Verbal and analytical strengths; data-focused; conscientiousness
Healthcare/Clinical Roles	Very Good	Medical knowledge capacity; emotional stability; conscientiousness; patient care aptitude
Project Management (Tech/Research)	Very Good	Conscientiousness, organization, analytical ability
Data Science/Analytics	Very Good	Cognitive profile, attention to detail, methodological rigor
Quality Assurance/Compliance	Good	Conscientiousness, detail-oriented, systematic
Technical Writing	Good	Bilingual, verbal ability, organized communication
Executive/C-Suite	Moderate	Conscientiousness supports; introversion requires intentional leadership development
Sales/Business Development	Moderate	Moderate extraversion; can develop skills but not natural fit

DOMAIN	FIT LEVEL	RATIONALE
Creative/Design Fields	Lower	Average openness; analytical strengths better deployed elsewhere
Pure Entrepreneurship	Lower	Conservative risk tolerance; stable organizational environment better fit

5.5 Development Areas and Recommendations

AREA	CURRENT STRENGTH	DEVELOPMENT OPPORTUNITY	STRATEGY
Communication	Verbal ability present	Public speaking/presentation skills	Toastmasters or executive communication coaching
Leadership	Conscientiousness, analysis	Interpersonal/emotional intelligence	Leadership development programs; mentorship
Networking	Introverted tendency	Professional relationship building	Small-group engagement; online networking; authenticity-based approach
Creativity	Average-to-above average	Innovation in structured domains	Design thinking workshops; cross-functional collaboration
Risk Assessment	Balanced (good foundation)	Strategic risk-taking	Business strategy courses; mentored entrepreneurship exposure

5.6 Career Path Recommendations

Optimal Career Trajectory:

- 1. Early Career (Age 25-35): Deep expertise development
  o Focus on technical mastery in chosen domain
  o Specialized certifications/advanced degrees (suits analytical profile)

o Build reputation as "expert" contributor

- 2. Mid-Career (Age 35-45): Leadership transition
  o Gradually increase team leadership responsibility
  o Develop mentoring relationships
  o Move toward specialist leadership roles (technical leadership, research direction)

- 3. Late Career (Age 45+): Strategic/Senior Role
  o Scientific leadership, research direction, policy influence
  o Potential institutional leadership with appropriate support
  o Knowledge transfer and legacy building

Recommended Career Paths:
● Principal Investigator in biomedical research
● Chief Scientific Officer in biotech/genomics company
● Director of Genomics/Precision Medicine in healthcare system
● Clinical Geneticist or Medical Director in genetic testing company
● Senior Data Scientist in healthcare/pharma

Clinical Genetics Interpretation Summary

Overall Risk Assessment

RISK CATEGORY	STATUS	ACTION REQUIRED
Cancer Screening	Elevated CRC Risk	Priority: Colonoscopy age 40
Neurological Health	Elevated AD Risk	Priority: Cognitive monitoring; prevention
Cardiometabolic Health	Intermediate Risk	Monitor: Lipids, glucose, BP; lifestyle
Longevity	Above Average	Favorable telomere genetics; maintain wellness
Career Fit	Excellent in biomedical/research fields	Aligned with background and genetics

Recommendations and Action Plan

Immediate Actions (Next 3 Months)

- 1. Schedule colonoscopy - Referral to gastroenterology for baseline colonoscopy (age-risk appropriate: age 40 vs. standard 45-50)
- 2. Baseline metabolic panel - Fasting glucose, lipid panel, liver/kidney function
- 3. Cardiovascular screening - Blood pressure monitoring; consider baseline EKG and CAC score at age 55
- 4. Cognitive baseline - Consider Montreal Cognitive Assessment (MoCA) or similar battery for baseline

Short-Term Plan (3-12 Months)

- 5. Lifestyle optimization:
  o Enroll in structured exercise program (150+ min/week moderate aerobic activity)
  o Begin Mediterranean or MIND diet protocol
  o Address sleep hygiene; consider sleep study if snoring/daytime sleepiness
- 6. Metabolic management:
  o Weight loss if overweight (targeting BMI <25); FTO heterozygosity responds well to structured interventions
  o Nutritionist consultation for personalized dietary guidance
- 7. Cognitive health interventions:
  o Engage in cognitive training activities (Lumosity, dual n-back)
  o Pursue intellectually engaging hobbies
  o Consider bilingual language learning or new skill acquisition

Ongoing Management (Annual/Biannual)

- 8. Annual screening:
  o Fasting metabolic panel (glucose, lipids, liver/kidney function)
  o Blood pressure monitoring
  o Weight/BMI assessment
  o Colonoscopy every 7-10 years if negative findings
- 9. Biennial assessments:
  o Cognitive function screening (age 60+)
  o Consider biomarker testing (tau-PET, amyloid-PET, blood phospho-tau) at age 65-70

- 10. Medication review:
  o Aspirin for primary prevention unlikely indicated given current risk profile
  o Lipid-lowering therapy (statin) may be considered based on 10-year ASCVD risk; discuss with physician

Limitations and Important Caveats

- 1. Multifactorial diseases: Complex diseases result from gene-environment interactions. Genetic predisposition is NOT determinism. Lifestyle modifications can substantially alter expressed phenotype.
- 2. Population specificity: This report uses primarily European and South Asian ancestry reference data. Some effect sizes may differ in other populations.
- 3. Rapidly evolving field: Genetic research continues to evolve. New discoveries may refine these interpretations. Regular review recommended (every 3-5 years).
- 4. Carrier screening: This report does not include comprehensive carrier screening for recessive genetic conditions. Separate carrier panel recommended if family planning is anticipated.
- 5. Rare variants: This report focuses on common genetic variants (SNPs, common CNVs). Rare pathogenic variants may be present and not fully reported here.
- 6. Privacy and ethics: Genetic information is sensitive. Recommend secure storage and careful consideration regarding sharing with family members.
- 7. Clinical validation: Genetic findings should be discussed with qualified genetic counselor or physician specializing in genomic medicine before clinical implementation.

References

[1] Raghavachari, N., et al. (2020). The impact of apolipoprotein E genetic variability in health and disease. Journal of Alzheimer's Disease, 72(3), 891-917.
[2] Ajnakina, O., et al. (2023). Polygenic propensity for longevity, APOE-ε4 status, and all-cause mortality. The Journals of Gerontology Series A.
[3] Codd, V., et al. (2013). Identification of seven loci affecting mean telomere length. Nature Genetics.
[4] Poosri, S., et al. (2024). Association of FTO variants with obesity and metabolic syndrome. Scientific Reports.

Appendix A: Technical Specifications

Sequencing Parameters:

- Platform: Illumina NovaSeq 6000
- Read length: 150 bp paired-end
- Target depth: 30×
- Quality threshold: Q ≥ 30 for 99.8% of bases

Bioinformatic Pipeline:

- Alignment: BWA-MEM (GRCh38/hg38)
- Variant calling: GATK HaplotypeCaller v4.2
- Annotation: VEP (Ensembl 110)
- CNV detection: MANTA/Canvas

Quality Assurance:

- Sample contamination: <1% (CONTAMINATOR)
- Genetic sex confirmation: XY (male)
- Relatedness check: No unexpected relatedness detected
- Hardy-Weinberg equilibrium: Pass

Appendix B: Additional Resources

Patient Education:
● National Institutes of Health (NIH) Genetics Home Reference: https://ghr.nlm.nih.gov/
● American College of Medical Genetics (ACMG): https://www.acmg.net/
● Genetic Alliance: https://www.geneticalliance.org/

Cancer Screening Guidelines:
● American Cancer Society: https://www.cancer.org/
● US Preventive Services Task Force (USPSTF): https://www.uspreventiveservicestaskforce.org/

Alzheimer's Disease Resources:
● Alzheimer's Association: https://www.alz.org/
● Brain Health Registry: https://www.brainHealthRegistry.org/

● ClinicalTrials.gov: https://clinicaltrials.gov/ (search for trials relevant to identified genetic risks)

Report Certification

This whole genome sequencing report has been interpreted according to current clinical genomics standards (ACMG/AMP guidelines for variant classification). Variants are classified as pathogenic, likely pathogenic, variant of uncertain significance (VUS), likely benign, or benign based on established criteria.

Report Certification: This report was analyzed and reviewed by:

Signed

Dr.[redacted]
Board-Certified Clinical Geneticist

[redacted]

Laboratory Director Approval:

Signed

Dr.[redacted]
Clinical Laboratory Director
Chiranjiv Genomics

[redacted]

Confidential - For Medical Use Only

This report contains sensitive personal health information. Unauthorized disclosure is prohibited.

Report generated: November 26, 2025, 12:48 PM IST
Document version: 1.0
Sample barcode: CHG-SAM-2025-087342